

INTRODUCTION:

- ▶ Li-Fi (Light Fidelity) uses light to transmit data.
- ▶ It is faster, secure and uses LED light for communication.
- ▶ This project transfers audio using LED light.

What is Li-Fi?

- ▶ Li-Fi means data transfer using visible light
- ▶ Invented by Prof. Harald Haas. Uses LED, photodiode, transmitter, receiver.

PROBLEM STATEMENT:

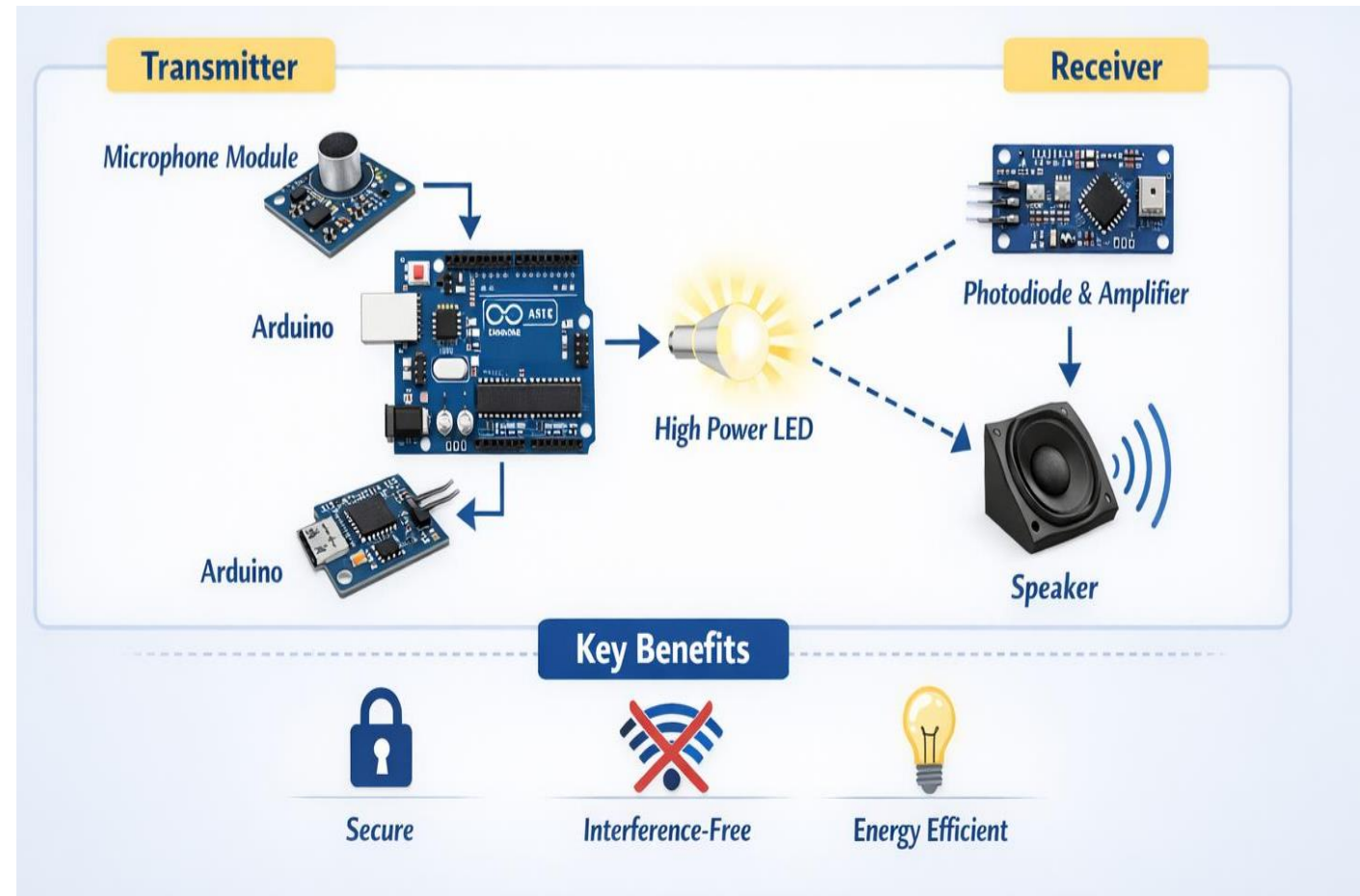
- ▶ Wi-Fi has security issues.
- ▶ Radio frequency congestion.
- ▶ Need safe & fast data transfer.

OBJECTIVE:

- ▶ To transfer audio using LED light.
- ▶ To control lighting smartly.
- ▶ To create secure communication.

BLOCK DIAGRAM:

This project demonstrates Li-fi technology where audio data is transmitted through the LED light using rapid intensity modulation. A microphone captures the sound which is sent via smart LED. A photodiode at the receiver detects the light signals, and the audio is amplified and played through a speaker.



COMPONENTS:

- ▶ LED
- ▶ Photodiode
- ▶ Arduino
- ▶ Audio amplifier
- ▶ Speaker
- ▶ Power supply
- ▶ Resistors, wires

WORKING PRINCIPLE:

- ▶ Audio Converted to electrical signal.
- ▶ Signal modulates LED light.
- ▶ Photodiode receives light.
- ▶ Signal converted back to audio.

APPLICATIONS:

- ▶ Smart homes
- ▶ Hospitals
- ▶ Airports
- ▶ Defense communication
- ▶ Smart classrooms

ADVANTAGES:

- ▶ High Speed
- ▶ No RF interference
- ▶ Secure
- ▶ Low cost
- ▶ Energy efficient

FUTURE SCOPE:

- ▶ Video transmission
- ▶ IoT integration
- ▶ Smart City application
- ▶ High-speed Li-Fi networks

CONCLUSION:

- ▶ Li-Fi is future technology.
- ▶ Smart lighting with audio transfer is efficient.
- ▶ Safe and secure communication system.



THANK YOU